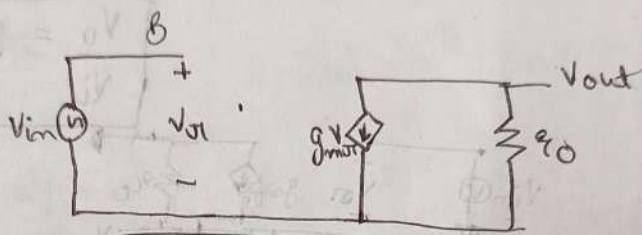
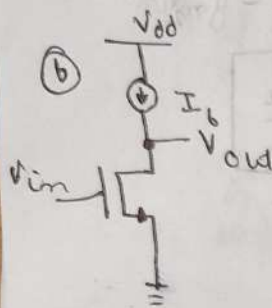
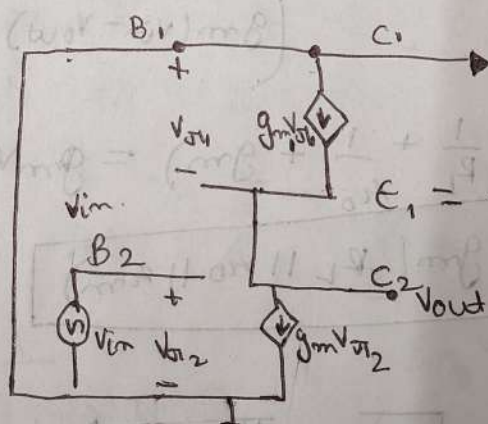
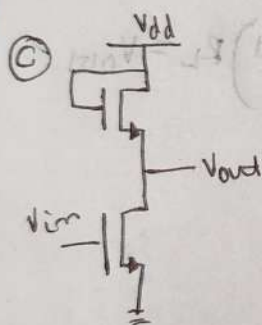


$$V_{out} = -g_m V_{gs} R_L = -g_m R_L V_{in}$$

$$\frac{V_{out}}{V_{in}} = -g_m R_L$$



$$\frac{V_{out}}{V_{in}} = -g_m r_o$$



$$g_m V_{gs1} = g_m V_{gs2}$$

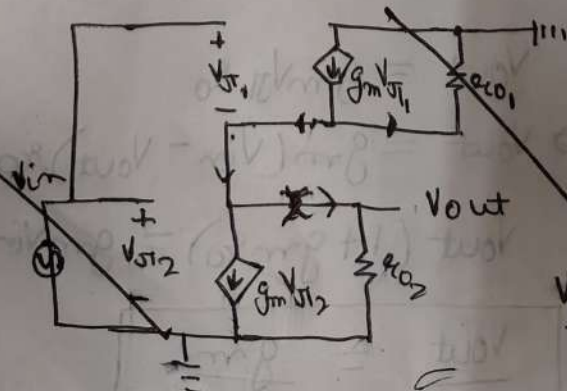
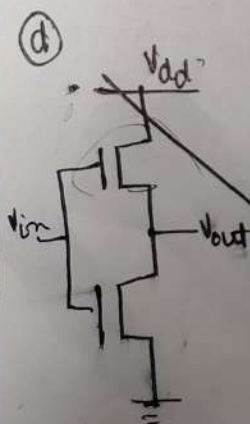
$$-g_m V_{out} = g_m V_{in}$$

$$\frac{V_{out}}{V_{in}} = -\frac{g_m}{g_m}$$

$$V_{in} = V_{gs2}$$

$$V_{out} = -V_{gs1}$$

$$\frac{V_{out}}{V_{in}} = -\frac{V_{gs1}}{V_{gs2}}$$



$$\frac{V_{out}}{r_{o2}} + g_m V_{in} = 0$$

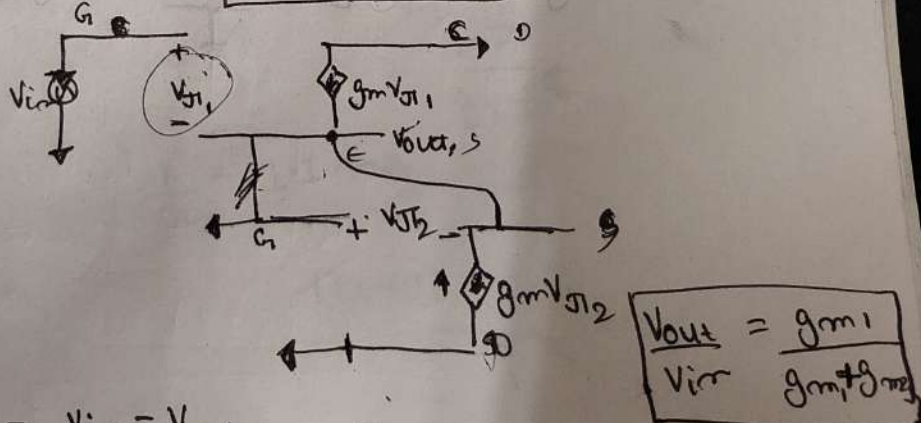
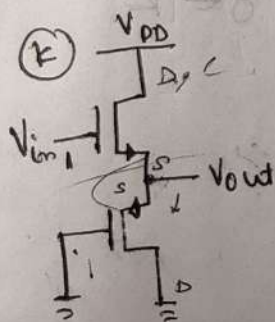
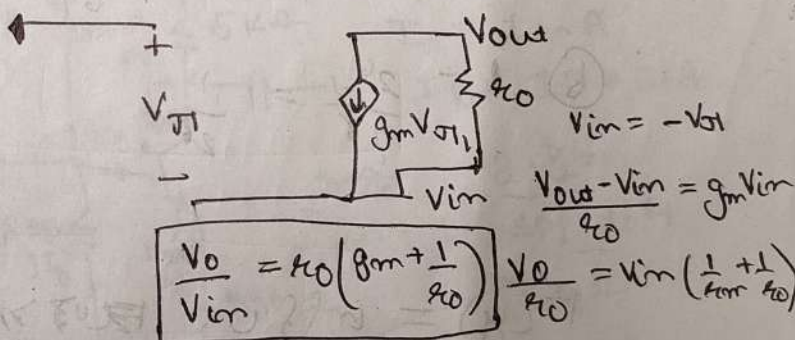
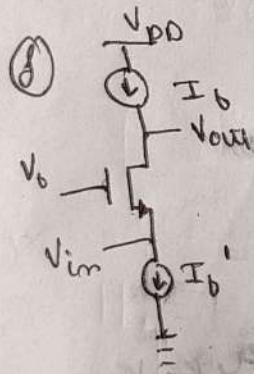
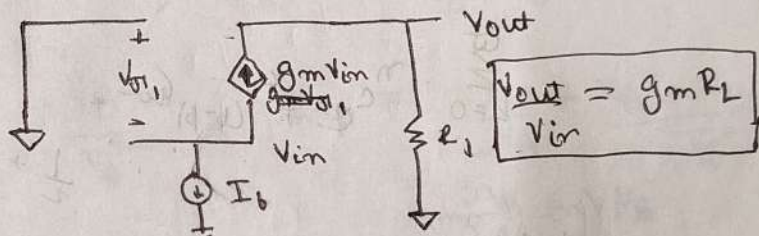
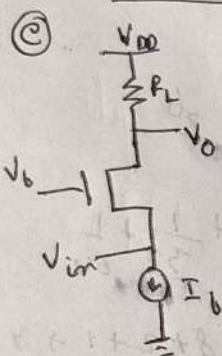
$$V_{GS1} = V_{in} - V_{out}$$

$$g_{m1}V_{GS1} - \frac{V_{out}}{r_{o1}} - \frac{V_{out}}{r_{o2}} - g_{m2}V_{GS2} = 0$$

$$g_{m1}(V_{in} - V_{out}) - V_{out} \left( \frac{1}{r_{o1}} + \frac{1}{r_{o2}} \right) - g_{m2}V_{in} = 0$$

$$V_{in}(g_{m1} - g_{m2}) = V_{out} \left( \frac{1}{r_{o1}} + \frac{1}{r_{o2}} + \frac{1}{r_{m1}} \right)$$

$$\frac{V_{out}}{V_{in}} = (g_{m1} - g_{m2}) (r_{o1} || r_{o2} || r_{m1})$$



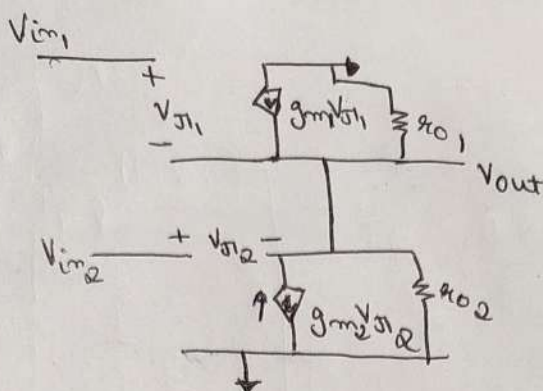
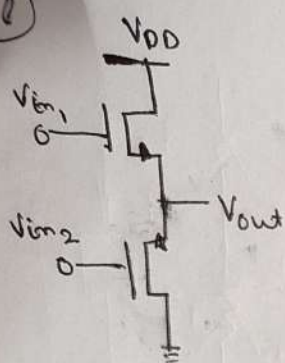
$$V_{GS1} = V_{in} - V_{out}$$

$$-V_{GS2} = V_{out}$$

$$-g_{m2}V_{GS2} = g_{m1}(V_{GS1})$$

$$\Rightarrow V_{in} g_{m1} = V_{out} (g_{m1} + g_{m2})$$

②



$$V_{gs2} = V_{in2} - V_o$$

$$V_{gs1} = V_{in1} - V_o$$

$$g_{m1}(V_{in1} - V_o) - \frac{V_o}{r_{o1}} = -g_{m2}(V_{in2} - V_o) + \frac{V_o}{r_{o2}}$$

$$V_{in1} g_{m1} + V_{in2} g_{m2} = V_o \left( \frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m1} + g_{m2} \right)$$

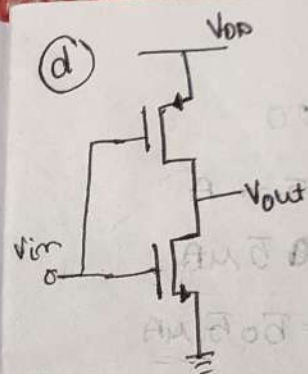
Gain w.r.t  $V_{in1} \Rightarrow$

$$\frac{V_o}{V_{in1}} = \frac{g_{m1}}{\frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m1} + g_{m2}}$$

Gain w.r.t  $V_{in2} \Rightarrow$

$$\frac{V_o}{V_{in2}} = \frac{g_{m2}}{\frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m1} + g_{m2}}$$

④



$$\Rightarrow V_o \left( \frac{1}{r_{o1}} + \frac{1}{r_{o2}} + g_{m1} + g_{m2} \right)$$

$$\left[ \frac{V_o}{V_{in}} \right]$$

$$r_{o1}, V_{out}$$

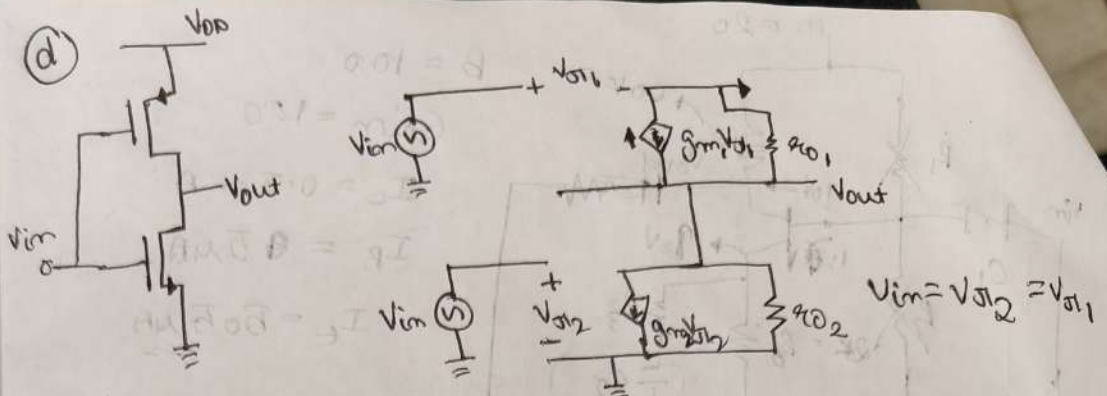
$$r_{o2}$$

$$) + \frac{V_o}{r_{o2}}$$

$$+ g_{m1} + g_{m2})$$

$$\frac{g_{m1} + g_{m2}}{g_{m1} + g_{m2}}$$

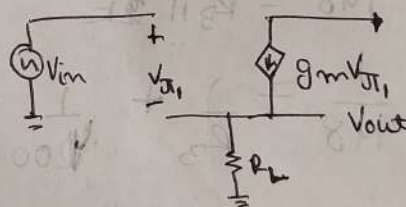
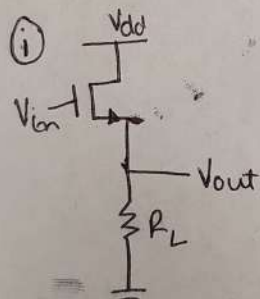
$$\frac{1 + g_{m1} + g_{m2}}{r_{o2}}$$



$$-\frac{V_o}{r_{o1}} + g_{m1}V_{in} = g_{m2}V_{in} + \frac{V_o}{r_{o2}}$$

$$\Rightarrow V_o \left( \frac{1}{r_{o1}} + \frac{1}{r_{o2}} \right) = -V_{in} (g_{m1} + g_{m2})$$

$$\boxed{\frac{V_o}{V_{in}} = -(g_{m1} + g_{m2})(r_{o1} || r_{o2})}$$

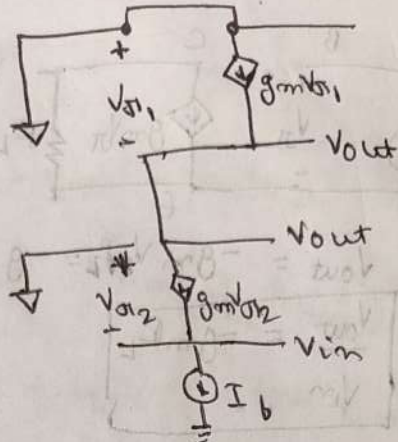
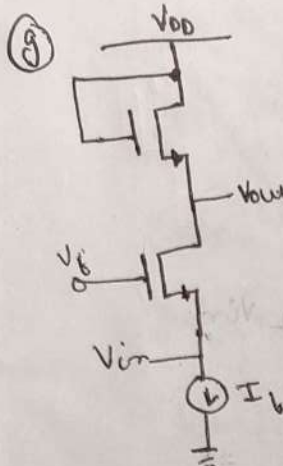


$$V_{gs1} = V_{in} - V_{out}$$

$$V_{out} = (V_{in} - V_{out}) g_{m1} R_L$$

$$\Rightarrow V_{out}(1 + g_{m1} R_L) = V_{in} g_{m1} R_L$$

$$\boxed{\frac{V_{out}}{V_{in}} = \frac{g_{m1} R_L}{1 + g_{m1} R_L}}$$

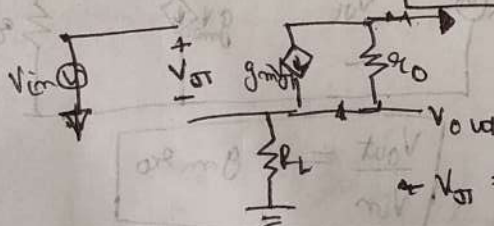
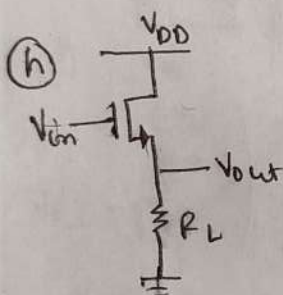


$$-V_{gs1} = V_{out}$$

$$-g_{m1}V_{out}$$

$$-g_{m1}V_0 = -g_{m2}V_{in}$$

$$\frac{V_0}{V_i} = \frac{g_{m2}}{g_{m1}}$$

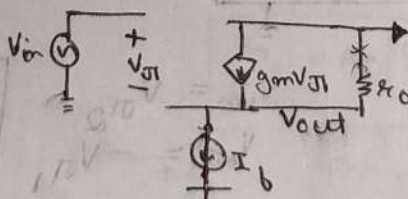
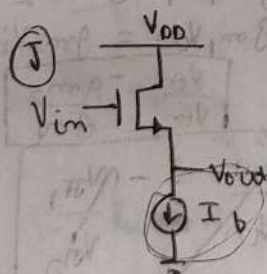


$$V_{gs} = V_{in} - V_{out}$$

$$\left( g_m(V_{in} - V_{out}) - \frac{V_{out}}{r_{o2}} \right) R_L = V_{out}$$

$$V_{out} \left( \frac{1}{R_L} + \frac{1}{r_{o2}} + g_m \right) = g_m V_{in}$$

$$\frac{V_{out}}{V_{in}} = g_m (R_L \parallel r_{o2} \parallel r_{o1})$$



$$V_{gs} = V_{in} - V_{out}$$

$$V_{out} = g_m V_{gs} r_o$$

$$\Rightarrow V_{out} = g_m (V_{in} - V_{out}) r_o$$

$$V_{out} (1 + g_m r_o) = g_m V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m r_o}{1 + g_m r_o}$$